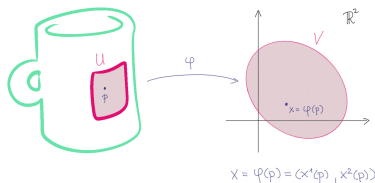


# Computing CMB temperature fluctuations for spherical spaces

Adriel Matei, Béla Schneider, Javier Vela, Juš Kocutar

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# Manifolds & Homotopy Groups



**Figure:** The prototypical example of a manifold a mug [13].

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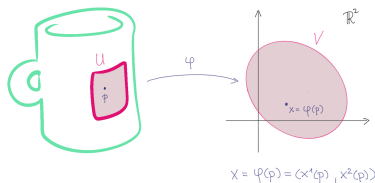


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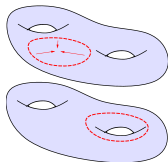


Figure: Two double tori with (non)-contractible paths [7].

# Quotients of the 3-sphere

- $SO(4)$  is isomorphic to the isometry group of  $\mathbb{S}^3$ .
- Subgroups  $\Gamma \leq SO(4)$  define **equivalence classes** of orbits on  $\mathbb{S}^3$  by the standard action of  $SO(4)$  on  $\mathbb{R}^4$ , i.e.  $x \sim y$  iff  $x = My$  for some  $M \in \Gamma$ .

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- The obtained space is (*sometimes*) a manifold. In particular, it is well defined and spherical for finite  $\Gamma$ .
- This can be easily generalized to the  $n$ -sphere.

# Lens Spaces

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## Definition (Lens space)

Given coprime  $p, z \in \mathbb{Z}$ , we define

$$L(p, q) := \mathbb{S}^3 / \langle M \rangle,$$

where  $\langle M \rangle$  is the group generated by

$$M := \begin{bmatrix} \cos(2\pi/p) & \sin(-2\pi/p) & 0 & 0 \\ \sin(2\pi/p) & \cos(2\pi/p) & 0 & 0 \\ 0 & 0 & \cos(2\pi q/p) & \sin(-2\pi q/p) \\ 0 & 0 & \sin(2\pi q/p) & \cos(2\pi q/p) \end{bmatrix}.$$

# The Laplace-Beltrami operator

## Definition (Laplace-Beltrami operator)

a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f.$$

- 1 The spectrum of this operator is a subset of  $[0, \infty)$ . We call it the *spectrum* of the manifold.



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- 1 The spectrum of this operator is a subset of  $[0, \infty)$ . We call it the *spectrum* of the manifold.
- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

# The silver lining

- Isospectral pairs are intriguing in their own right, but they cannot occur in dimension 3.
- We can thus attempt to infer the shape of our universe based on its spectrum.

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- The set of all possible wave numbers [for which a nonzero solution exists] depends on  $\Gamma$  - usually a proper subset of  $\mathbb{N}$ .
- Solutions to the Helmholtz equation are calculated for each  $\Gamma$  and CMB anisotropy  $\frac{\delta T}{T}$  computed as a sum of spherical harmonics.



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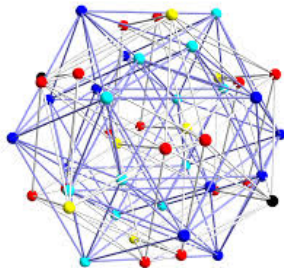


Figure: Graphical representation of the binary tetrahedral group [5]

# Inhomogeneous spherical space

- Manifolds of the form  $\mathbb{S}^3/\Gamma$  with  $\Gamma$  a group acting on  $\mathbb{S}^3$ .
- **Multi-connected** space: it has non-contractable loops.
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  - ① homogeneous:  $N3$  and  $L(8, 1)$ .
  - ② inhomogeneous:  $N2 \equiv L(8, 3)$ .
- Results: inhomogeneous spaces have **more variety** in the CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is **observer dependent**.



# Statistical Isotropy and Hypothesis

## Theoretical Expectation:

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**Key Question:** - Does the observed CMB data deviate from statistical isotropy? - If  $\langle a_{\ell m} \rangle \neq 0$ , this suggests a preferred cosmic direction.

# Methodology: Sky Masking and the Test Statistic

## Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
- A **sky mask function**  $A(\hat{n})$  removes contaminated regions, but introduces bias.

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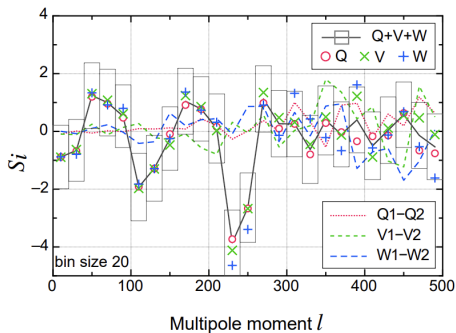
## Monte Carlo Simulations:

- We generate thousands of **simulated CMB skies** assuming isotropy.
- Each sky has different random  $a_{\ell m}$ , drawn from:

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell)$$

- The observed WMAP data is compared against these simulations.

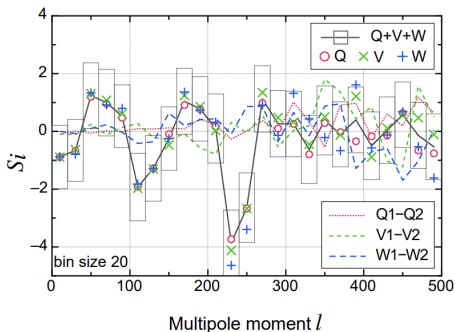
# Results and Interpretation



**Figure:** The decorrelated band mean test statistic values over multipole ranges [9].



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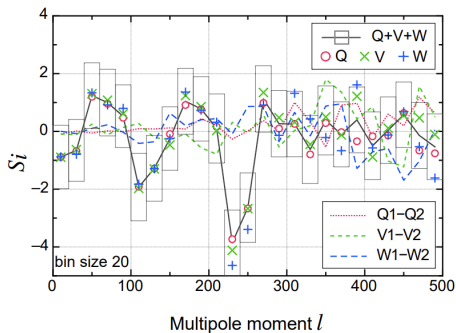


## Key Findings:

- A statistically significant deviation was found in  $221 \leq \ell \leq 240$ .
- This suggests a potential **preferred cosmic direction**.

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## Key Findings:

- A statistically significant deviation was found in  $221 \leq \ell \leq 240$ .
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## Possible Explanations:

- A real cosmological signal? → A finite universe or new physics.
- A systematic effect? → Foreground contamination or instrumental noise.

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- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.
- ④ Statistical test results suggest possibilities of **finite multi-connected** topology.

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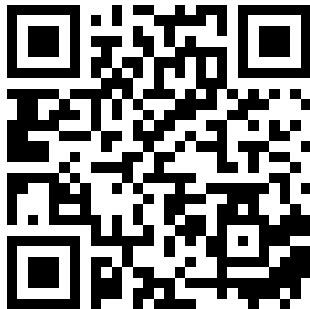


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💫 Thank You! 💫